

Financial Mathematics Practice Worksheet

Chapter 11.2 – Continuous Compounding (Complete Set: Tasks 1–20)

Tasks 1–20, sorted from easiest to hardest (difficulty 1/5 to 5/5). Key formulas: with continuous compounding of interest at nominal annual rate r , an initial principal S_0 grows after t years to $S(t) = S_0 \cdot e^{rt}$; each year the principal is multiplied by the same fixed factor e^r , so $S(t+1) = S(t) \cdot e^r$. The effective interest rate with continuous compounding at annual rate r is $e^r - 1$, and this is also the largest effective annual rate obtainable at nominal rate r under any compounding frequency whatsoever, since $(1 + r/n)^n \rightarrow e^r$ as $n \rightarrow \infty$. Continuous compounding is therefore always at least as profitable for the lender as any less-frequent compounding at the SAME nominal rate, and the gap between them grows larger as either the interest rate or the number of years increases (and stays small for low rates over short periods) — though a higher nominal rate under less-frequent compounding can still outperform a lower nominal rate compounded continuously, since the ceiling result only compares schedules sharing one common rate. The identical exponential law, run with a negative rate, describes continuous depreciation: a value shrinking continuously at annual rate δ follows $v(t) = v_0 \cdot e^{-\delta t}$. Tasks 11–20 additionally combine these same results in multi-step ways — solving for an unknown rate or time from observed growth or decay, chaining sequential compounding phases, and comparing several instruments or assets at once — but every task relies exclusively on the formulas above. Round dollar amounts to 2 decimal places and rates to 2 decimal places unless otherwise noted.

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20 Capstone: A Three-Asset Family Office Portfolio Under Continuous Growth and Decay

5.0/5

Task 1. Continuous Compounding on a Bakery's Business Account

Difficulty: 1.0/5

Ms. Delgado, the owner of an artisan bakery, deposits \$4,500 into a business savings account that compounds interest continuously at a nominal annual rate of 5%.

Statements (True / False):

- a) The balance after exactly 1 year is \$4,730.72.
- b) The interest earned during the first year is \$230.72.
- c) If the bank compounded the same 5% nominal rate annually (once per year) instead of continuously, the year-end balance would be \$4,735.00.
- d) The dollar difference between continuous compounding and annual compounding at the same nominal rate is \$5.72.
- e) The growth factor $e^{0.05}$ rounds to 1.0400.

Solution – Task 1

Given:

- $S_0 = \$4,500$
- Nominal annual rate $r = 5\% = 0.05$
- Compounding: continuous
- $t = 1$ year

Formula(s):

- $S(t) = S_0 \cdot e^{(rt)}$

Step-by-Step Calculations:

$$S(1) = 4,500 \times e^{0.05} = 4,500 \times 1.0512711 = \$4,730.72.$$

$$\text{Interest} = 4,730.72 - 4,500.00 = \$230.72.$$

$$\text{Annual (m = 1) compounding at 5\%: } S = 4,500 \times 1.05 = \$4,725.00.$$

$$\text{Difference} = 4,730.72 - 4,725.00 = \$5.72.$$

$$e^{0.05} = 1.0512711, \text{ which rounds to } 1.0513, \text{ not } 1.0400.$$

Explanation of Each Statement:

a) TRUE. Plugging $S_0 = 4,500$, $r = 0.05$, and $t = 1$ into $S(t) = S_0 e^{(rt)}$ gives $4,500 \times e^{0.05} = 4,500 \times 1.0512711 = \$4,730.72$, matching the statement exactly. Continuous compounding assumes interest is being added at every instant rather than at discrete intervals, and this exponential formula captures that limiting behavior directly.

b) TRUE. Interest earned is simply the year-end balance minus the original deposit: $4,730.72 - 4,500.00 = \$230.72$, matching the statement exactly.

c) FALSE. Annual compounding at 5% for one year gives $4,500 \times 1.05 = \$4,725.00$, not \$4,735.00. The stated figure overstates the true annual-compounding balance by \$10.00, and in any case it would still fall short of the continuous-compounding balance of \$4,730.72, since continuous compounding is the most frequent schedule possible and can never be beaten by a less-frequent one at the same nominal rate.

d) TRUE. The correctly computed annual balance is \$4,725.00, so the gap versus the continuous balance is $4,730.72 - 4,725.00 = \$5.72$, exactly as stated.

e) FALSE. e raised to the power 0.05 equals approximately 1.0512711, which rounds to 1.0513, not 1.0400 — the figure of 1.0400 would instead correspond to a much smaller exponent and does not describe this 5% continuously-compounded growth factor at all.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 2. Multi-Year Continuous Compounding for a Coffee Roasting Company

Difficulty: 1.2/5

A specialty coffee roasting company invests \$3,200 of retained earnings into a fund that compounds continuously at a nominal annual rate of 8%, and plans to leave the funds untouched for 6 years.

Statements (True / False):

- a) After 6 years, the balance is approximately \$5,171.44.
- b) Doubling the 3-year balance gives the correct 6-year balance.
- c) The total interest earned over the 6 years is approximately \$2,000.00.
- d) The 12-year balance would be exactly double the 6-year balance.
- e) The 12-year balance is less than double the 6-year balance.

Solution – Task 2

Given:

- $S_0 = \$3,200$
- Nominal annual rate $r = 8\% = 0.08$
- Compounding: continuous
- $t = 6$ years (3 and 12 years for parts b, d, e)

Formula(s):

- $S(t) = S_0 \cdot e^{rt}$

Step-by-Step Calculations:

$$S(6) = 3,200 \times e^{(0.08 \times 6)} = 3,200 \times e^{0.48} = 3,200 \times 1.6161681 = \$5,171.44.$$

$$S(3) = 3,200 \times e^{(0.08 \times 3)} = 3,200 \times e^{0.24} = 3,200 \times 1.2712492 = \$4,067.998 \approx \$4,068.00; \text{ doubling gives } \$8,135.99, \text{ which does not equal } S(6).$$

$$\text{Interest} = 5,171.44 - 3,200.00 = \$1,971.44, \text{ not } \$2,000.00.$$

$$S(12) = 3,200 \times e^{(0.08 \times 12)} = 3,200 \times e^{0.96} = 3,200 \times 2.6116965 = \$8,357.43.$$

$$\text{Double of } S(6) = 2 \times 5,171.44 = \$10,342.88; \text{ since } \$8,357.43 < \$10,342.88, \text{ the 12-year balance is indeed less than double the 6-year balance.}$$

Explanation of Each Statement:

- a) TRUE.** $S(6) = 3,200 \times e^{0.48} = \$5,171.44$, matching the statement exactly.
- b) FALSE.** The 3-year balance of \$4,068.00 is correctly computed, but doubling it to get \$8,135.99 does not reproduce the true 6-year balance of \$5,171.44. Continuous growth compounds multiplicatively rather than additively, so extending the holding period from 3 to 6 years squares the 3-year growth factor (since $e^{0.48} = (e^{0.24})^2$) rather than simply doubling the balance, and the true 6-year figure is well below the doubled estimate.
- c) FALSE.** The actual interest earned is $5,171.44 - 3,200.00 = \$1,971.44$, not \$2,000.00 — the stated figure is a rounded-up approximation that does not match the precise calculation.
- d) FALSE.** The true 12-year balance is $3,200 \times e^{0.96} = \$8,357.43$, which is noticeably less than double the 6-year balance of \$5,171.44 (which would be \$10,342.88). As in part (b), extending the time span multiplies growth factors rather than balances, so doubling the number of years does not double the account value.
- e) TRUE.** The correctly computed 12-year balance is \$8,357.43, and comparing this to double the 6-year balance (\$10,342.88) confirms that \$8,357.43 is indeed smaller, consistent with the multiplicative (rather than additive) nature of continuous exponential growth.

Final Answers — a: TRUE b: FALSE c: FALSE d: FALSE e: TRUE

Task 3. Effective Annual Rate on a Continuously-Compounded Bond Fund

Difficulty: 1.4/5

An asset manager quotes a bond fund's nominal annual rate of 9%, compounded continuously, and a client asks how the effective annual yield (EAR) compares to the nominal rate, and how it would change if the nominal rate doubled to 18%.

Statements (True / False):

- a) The effective annual rate (EAR) is approximately 9.42%.
- b) On a \$15,000 investment, the year-end balance is \$16,412.61.
- c) The EAR exceeds the nominal rate by more than 0.75 percentage points.
- d) If the nominal rate doubled to 18%, the resulting EAR would exceed double the original EAR.
- e) At the 18% nominal rate, the EAR exceeds 19.5%.

Solution – Task 3

Given:

- Nominal annual rate $r = 9\% = 0.09$ (and 18% for parts d, e)
- Compounding: continuous
- $P = \$15,000$ (part b)

Formula(s):

- $\text{EAR} = e^r - 1$
- $S(t) = S_0 \cdot e^{(rt)}$

Step-by-Step Calculations:

$$\text{EAR} = e^{0.09} - 1 = 1.0941743 - 1 = 0.0941743 \approx 9.42\%.$$

$$S(1) = 15,000 \times e^{0.09} = 15,000 \times 1.0941743 = \$16,412.61.$$

$$\text{Difference from nominal} = 9.4174\% - 9.00\% = 0.4174 \text{ percentage points.}$$

$$\text{At 18\%: } \text{EAR} = e^{0.18} - 1 = 1.1972174 - 1 = 0.1972174 \approx 19.72\%.$$

Double of the original EAR = $2 \times 9.4174\% = 18.8349\%$; since $19.72\% > 18.83\%$, the new EAR does exceed double the original.

Explanation of Each Statement:

- a) TRUE. $\text{EAR} = e^{0.09} - 1 \approx 9.42\%$, matching the statement exactly.
- b) TRUE. $S(1) = 15,000 \times e^{0.09} = \$16,412.61$, matching the statement exactly.
- c) FALSE. The actual gap between the EAR and the nominal rate is only about 0.4174 percentage points ($9.4174\% - 9.00\%$), which is well under 0.75 percentage points, not more than it.
- d) TRUE. Doubling the nominal rate to 18% raises the EAR to approximately 19.72%, which is indeed higher than double the original EAR of 18.83%. This happens because e^r grows faster than linearly in r , so doubling the exponent more than doubles the resulting effective rate above the nominal baseline.
- e) TRUE. $e^{0.18} - 1 \approx 19.72\%$, which is above 19.5%, matching the statement exactly.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: TRUE

Task 4. Comparing Growth Factors on an Equipment Leasing Contract

Difficulty: 1.6/5

An equipment leasing firm structures a financing contract at a nominal annual rate of 10%, and wants to compare the growth factor K (the amount to which \$1 grows over one year) under yearly, semi-annual, and continuous compounding, applied to a \$75,000 balance.

Statements (True / False):

- a) Under yearly compounding, $K = 1.1000$.
- b) Under semi-annual compounding, $K = 1.1025$.
- c) Under continuous compounding, $K \approx 1.1052$.
- d) On the \$75,000 balance, continuous compounding exceeds semi-annual compounding by \$250.32.
- e) The gap between semi-annual and yearly compounding is larger than the gap between continuous and semi-annual compounding.

Solution – Task 4

Given:

- Nominal annual rate $r = i = 10\% = 0.10$
- Compounding schedules: yearly ($m = 1$), semi-annual ($m = 2$), continuous
- $P = \$75,000$ (parts d, e)

Formula(s):

- $K_{\text{yearly}} = 1 + i$
- $K_{\text{semi}} = (1 + i/2)^2$
- $K_{\text{continuous}} = e^r$

Step-by-Step Calculations:

$$K_{\text{yearly}} = 1 + 0.10 = 1.1000.$$

$$K_{\text{semi}} = (1.05)^2 = 1.1025.$$

$$K_{\text{cont}} = e^{0.10} = 1.1051709.$$

$$75,000 \times (K_{\text{cont}} - K_{\text{semi}}) = 75,000 \times (1.1051709 - 1.1025) = 75,000 \times 0.0026709 = \$200.32, \text{ not } \$250.32.$$

$$75,000 \times (K_{\text{semi}} - K_{\text{yearly}}) = 75,000 \times 0.0025 = \$187.50; \text{ comparing } 187.50 \text{ to } 200.32, \text{ the semi-vs-yearly gap is actually the smaller of the two.}$$

Explanation of Each Statement:

- a) TRUE. $K_{\text{yearly}} = 1 + 0.10 = 1.1000$, matching the statement exactly.
- b) TRUE. $K_{\text{semi}} = (1.05)^2 = 1.1025$, matching the statement exactly.
- c) TRUE. $K_{\text{cont}} = e^{0.10} \approx 1.1052$, matching the statement exactly.
- d) FALSE. The correctly computed gap on \$75,000 is $75,000 \times (1.1051709 - 1.1025) = \200.32 , not \$250.32; the stated figure overstates the true difference by exactly \$50.00.
- e) FALSE. The semi-annual-versus-yearly gap is $75,000 \times (1.1025 - 1.1000) = \187.50 , which is in fact smaller than the continuous-versus-semi-annual gap of \$200.32, not larger. This is consistent with the general pattern that each successive increase in compounding frequency adds a further boost, and as frequency approaches its continuous limit those incremental boosts do not need to shrink monotonically in absolute dollar terms.

Final Answers — a: TRUE b: TRUE c: TRUE d: FALSE e: FALSE

Task 5. Year-Over-Year Growth by a Fixed Factor in a College Endowment Sub-Fund

Difficulty: 1.8/5

A college endowment's growth sub-fund is currently valued at \$95,000 and earns continuous compounding at a nominal annual rate of 4.5%. The board wants to project the balance forward using the property that the principal is multiplied by a fixed factor each year.

Statements (True / False):

- a) One year from now, the balance will be approximately \$98,500.00.
- b) Two years from now, the balance will be approximately \$103,946.56.
- c) The dollar increase in year 1 is larger than the dollar increase in year 2.
- d) Each year, the balance is multiplied by a different factor.
- e) If the nominal rate were doubled to 9%, the year-over-year growth factor would also exactly double.

Solution – Task 5

Given:

- $S(0) = \$95,000$
- Nominal annual rate $r = 4.5\% = 0.045$ (and 9% for part e)
- Compounding: continuous

Formula(s):

- $S(t + 1) = S(t) \cdot e^r$
- $S(t) = S_0 \cdot e^{(rt)}$

Step-by-Step Calculations:

$S(1) = 95,000 \times e^{0.045} = 95,000 \times 1.0460279 = \$99,372.65$, not \$98,500.00.

$S(2) = 95,000 \times e^{0.09} = 95,000 \times 1.0941743 = \$103,946.56$.

Increase(0→1) = $99,372.65 - 95,000.00 = \$4,372.65$. Increase(1→2) = $103,946.56 - 99,372.65 = \$4,573.91$.

The textbook property $S(t + 1) = S(t) \cdot e^r$ establishes that the SAME factor $e^{0.045} \approx 1.0460$ is applied every year, regardless of the current balance.

At 9%: $e^{0.09} = 1.0941743$, which is far less than $2 \times e^{0.045} = 2.0920557$.

Explanation of Each Statement:

- a) FALSE.** $S(1) = 95,000 \times e^{0.045} = \$99,372.65$, not \$98,500.00 — the stated figure is too low by roughly \$872.65.
- b) TRUE.** $S(2) = 95,000 \times e^{0.09} = \$103,946.56$, matching the statement exactly.
- c) FALSE.** The increase actually grows larger over time as the balance compounds on itself: the year-1 increase is \$4,372.65, which is smaller than the year-2 increase of \$4,573.91, not larger as claimed.
- d) FALSE.** This directly contradicts the property established in the text, $S(t + 1) = S(t) \cdot e^r$: the multiplying factor e^r depends only on the fixed nominal rate r , not on the current size of the balance, so the SAME factor (here, $e^{0.045} \approx 1.0460$) is applied every single year no matter how large the balance has already grown.
- e) FALSE.** Doubling the rate does not double the growth factor, because $e^{(2r)}$ equals $(e^r)^2$, not $2 \times e^r$. Here $e^{0.09} \approx 1.0942$, which is nowhere near the claimed 2.0921 — that number instead comes from incorrectly doubling $e^{0.045}$ directly rather than doubling the exponent inside it.

Final Answers — a: FALSE b: TRUE c: FALSE d: FALSE e: FALSE

Task 6. Continuous Depreciation of a Courier Company's Van Fleet

Difficulty: 2.0/5

A courier company's van fleet has a combined initial value of \$60,000 and depreciates continuously at an annual rate of 10% ($\delta = 0.10$), so that its value after t years follows $v(t) = v_0 \cdot e^{(-\delta t)}$.

Statements (True / False):

- a) The fleet's value after 4 years is approximately \$40,219.20.
- b) The fleet's value after 7 years is approximately \$29,795.12.
- c) The 4-year value represents about 67.03% of the original \$60,000 value.
- d) If the depreciation rate were instead 20% (double 10%), the 4-year value would fall below \$25,000.
- e) The fleet's dollar decline during the first year alone is larger than its dollar decline during the fourth year alone.

Solution – Task 6

Given:

- $v_0 = \$60,000$
- Annual depreciation rate $\delta = 10\% = 0.10$ (and 20% for part d)
- Compounding: continuous

Formula(s):

- $v(t) = v_0 \cdot e^{(-\delta t)}$

Step-by-Step Calculations:

$$v(4) = 60,000 \times e^{(-0.10 \times 4)} = 60,000 \times e^{(-0.40)} = 60,000 \times 0.6703200 = \$40,219.20.$$

$$v(7) = 60,000 \times e^{(-0.10 \times 7)} = 60,000 \times e^{(-0.70)} = 60,000 \times 0.4965853 = \$29,795.12.$$

$$\text{Percentage of original} = 40,219.20 / 60,000 = 0.670320 \approx 67.03\%.$$

At $\delta = 0.20$: $v(4) = 60,000 \times e^{(-0.80)} = 60,000 \times 0.4493290 = \$26,959.74$, which is above \$25,000, not below it.

$$v(1) = 60,000 \times e^{(-0.10)} = \$54,290.25, \text{ so year-1 loss} = 60,000 - 54,290.25 = \$5,709.75. \\ v(3) = 60,000 \times e^{(-0.30)} = \$44,449.09, \text{ so year-4 loss} = v(3) - v(4) = 44,449.09 - 40,219.20 = \$4,229.89.$$

Explanation of Each Statement:

- a) TRUE. $v(4) = 60,000 \times e^{(-0.40)} = \$40,219.20$, matching the statement exactly.
- b) TRUE. $v(7) = 60,000 \times e^{(-0.70)} = \$29,795.12$, matching the statement exactly.
- c) TRUE. $40,219.20 / 60,000 = 0.670320$, which is approximately 67.03%, matching the statement exactly.
- d) FALSE. Doubling the depreciation rate to 20% gives $v(4) = 60,000 \times e^{(-0.80)} \approx \$26,959.74$, which remains above \$25,000 rather than falling below it.
- e) TRUE. Because continuous depreciation removes a fixed PROPORTION of the current value each year rather than a fixed dollar amount, the dollar decline is largest when the base value is largest (i.e., early on) and shrinks as the value itself shrinks. The first-year loss of about \$5,709.75 is indeed larger than the fourth-year loss of about \$4,229.89, confirming this pattern.

Final Answers — a: TRUE b: TRUE c: TRUE d: FALSE e: TRUE

Task 7. Doubling Time for a REIT's Continuously-Compounded Reserve Account

Difficulty: 2.3/5

A real estate investment trust (REIT) places \$18,000 into a reserve account earning continuous compounding at a nominal annual rate of 5.5%, and wants to know how long it will take the balance to double, and what happens after three such doubling periods.

Statements (True / False):

- a) The doubling time at 5.5% is approximately 12.60 years.
- b) At exactly 12.60 years, the balance reaches approximately \$36,000.00.
- c) If the rate were instead 11%, the doubling time would also be approximately 12.60 years, unchanged.
- d) After three full doubling periods, the balance would grow to 6 times the original principal.
- e) A higher interest rate r gives a longer doubling time.

Solution – Task 7

Given:

- $S_0 = \$18,000$
- Nominal annual rate $r = 5.5\% = 0.055$ (and 11% for part c)
- Compounding: continuous
- Target: $S(t) = 2S_0$

Formula(s):

- $S(t) = S_0 \cdot e^{(rt)}$
- Doubling time: $e^{(rt)} = 2 \Rightarrow t = \ln(2)/r$

Step-by-Step Calculations:

$$\ln(2) = 0.693147.$$

$$t = 0.693147 / 0.055 = 12.6027 \text{ years.}$$

At 11%: $t = 0.693147 / 0.11 = 6.3013$ years, which is half of 12.6027, not unchanged.

$$S(12.6027) = 18,000 \times e^{(0.055 \times 12.6027)} = 18,000 \times e^{0.693147} = 18,000 \times 2 = \$36,000.00.$$

After 3 doubling periods (37.8080 years), the growth factor is $2 \times 2 \times 2 = 8$, so $S = 18,000 \times 8 = \$144,000.00$, not $6 \times 18,000 = \$108,000.00$.

Explanation of Each Statement:

- a) TRUE.** $t = \ln(2)/0.055 \approx 12.60$ years, matching the statement exactly.
- b) TRUE.** By construction, $e^{(rt)} = 2$ at the doubling time, so the balance is exactly \$36,000.00, twice the original \$18,000, matching the statement exactly.
- c) FALSE.** Doubling time is inversely proportional to the rate, not independent of it — doubling the rate to 11% HALVES the doubling time to approximately 6.30 years, it does not leave it unchanged at 12.60 years.
- d) FALSE.** Each successive doubling period multiplies the balance by 2 again, so after three full doubling periods the balance has grown by a factor of $2 \times 2 \times 2 = 8$, not 6. Eight times the original \$18,000 principal is \$144,000.00, not the \$108,000.00 implied by a 6-times multiplier.
- e) FALSE.** Since $t = \ln(2)/r$ has r in the denominator, a LARGER r produces a SMALLER (shorter) doubling time, not a longer one — this is exactly why doubling the rate from 5.5% to 11% cut the doubling time in half rather than lengthening it.

Final Answers — a: TRUE b: TRUE c: FALSE d: FALSE e: FALSE

Task 8. Time for a Stamping Press to Lose Value Under Continuous Depreciation

Difficulty: 2.5/5

A manufacturing plant's stamping press has an initial value of \$120,000 and depreciates continuously at an annual rate of 18% ($\delta = 0.18$). Management wants to know how long it will take for the press to lose 60% of its original value (retaining only 40%), and how that compares to a slower depreciation scenario and to losing a larger share of value.

Statements (True / False):

- a) Setting $v_0 \cdot e^{(-\delta t)} = 0.40v_0$ and solving gives $t = \ln(2.5)/\delta$.
- b) The press will have lost 60% of its value after approximately 5.09 years.
- c) At that point, the press's remaining value is approximately \$48,000.00.
- d) If the depreciation rate were instead 9%, the time to lose 60% of value would double to approximately 10.18 years.
- e) The time to lose 80% of the value is longer than the time to lose 60% of the value.

Solution – Task 8

Given:

- $v_0 = \$120,000$
- Annual depreciation rate $\delta = 18\% = 0.18$ (and 9% for part d)
- Compounding: continuous
- Target: $v(t) = 0.40v_0$ (and $0.20v_0$ for part e)

Formula(s):

- $v(t) = v_0 \cdot e^{(-\delta t)}$
- $e^{(-\delta t)} = \text{fraction remaining} \Rightarrow t = -\ln(\text{fraction})/\delta = \ln(1/\text{fraction})/\delta$

Step-by-Step Calculations:

$$1/0.40 = 2.5, \text{ and } \ln(2.5) = 0.916291.$$

$$t = 0.916291 / 0.18 = 5.0905 \text{ years.}$$

$$v(5.0905) = 120,000 \times 0.40 = \$48,000.00.$$

At $\delta = 0.09$: $t = 0.916291 / 0.09 = 10.1810$ years, which is indeed double 5.0905 years, since halving δ doubles t .

For 80% loss (retain 20%): $1/0.20 = 5$, $\ln(5) = 1.609438$. $t = 1.609438 / 0.18 = 8.9413$ years, which is longer than 5.0905 years.

Explanation of Each Statement:

- a) TRUE.** Dividing both sides by v_0 and taking natural logs of $1/0.40 = 2.5$ gives $\delta t = \ln(2.5)$, so $t = \ln(2.5)/\delta$, matching the statement exactly.
- b) TRUE.** $t = \ln(2.5)/0.18 \approx 5.09$ years, matching the statement exactly.
- c) TRUE.** Retaining 40% of the original \$120,000 value gives $120,000 \times 0.40 = \$48,000.00$, matching the statement exactly.
- d) TRUE.** Since $t = \ln(2.5)/\delta$ has δ in the denominator, halving δ from 18% to 9% doubles the required time, from 5.09 years to 10.18 years, exactly as stated.
- e) TRUE.** Losing a larger share of the original value naturally takes longer under continuous proportional depreciation: reaching only 20% remaining (an 80% loss) requires $t = \ln(5)/0.18 \approx 8.94$ years, which is indeed longer than the 5.09 years needed to reach 40% remaining (a 60% loss).

Final Answers — a: TRUE b: TRUE c: TRUE d: TRUE e: TRUE

Task 9. How the Continuous-vs-Annual Gap Widens With Rate and Time at a Regional Bank

Difficulty: 2.8/5

A regional bank's marketing team wants to demonstrate to clients how the dollar gap between continuous compounding and ordinary annual compounding changes with the interest rate and the holding period. It compares nominal rates of 3% and 15%, each applied for 1 year to a \$25,000 deposit, and then extends the 3% case to an 8-year horizon.

Statements (True / False):

- a) At a 3% nominal rate over 1 year, the gap between continuous and annual compounding on \$25,000 is approximately \$11.36.
- b) At a 15% nominal rate over 1 year, the gap between continuous and annual compounding on \$25,000 is approximately \$295.86.
- c) The 1-year gap at 15% is more than 30 times larger than the 1-year gap at 3%.
- d) Extending the 3% comparison from 1 year to 8 years increases the dollar gap to approximately \$111.98.
- e) Continuous compounding becomes less advantageous to the lender over longer holding periods.

Solution – Task 9

Given:

- $P = \$25,000$
- Nominal annual rates $r = 3\%$ and $r = 15\%$
- Compounding: continuous vs. annual ($m = 1$)
- $t = 1$ year (and 8 years for part d, e)

Formula(s):

- $S_{\text{cont}} = P \cdot e^{(rt)}$
- $S_{\text{annual}} = P \cdot (1 + r)^t$

Step-by-Step Calculations:

At 3%, 1 yr: $S_{\text{cont}} = 25,000 \times e^{0.03} = \$25,761.36$. $S_{\text{annual}} = 25,000 \times 1.03 = \$25,750.00$. Gap = \$11.36.

At 15%, 1 yr: $S_{\text{cont}} = 25,000 \times e^{0.15} = \$29,045.86$. $S_{\text{annual}} = 25,000 \times 1.15 = \$28,750.00$. Gap = \$295.86.

Ratio = $295.86 / 11.36 \approx 26.04$, which is less than 30.

At 3%, 8 yrs: $S_{\text{cont}} = 25,000 \times e^{0.24} = \$31,781.23$. $S_{\text{annual}} = 25,000 \times (1.03)^8 = \$31,669.25$. Gap = \$111.98.

The 8-year gap (\$111.98) is far larger than the 1-year gap (\$11.36) at the same 3% rate, so the gap GROWS, not shrinks, as the horizon lengthens.

Explanation of Each Statement:

- a) TRUE. Gap = $25,761.36 - 25,750.00 = \11.36 , matching the statement exactly.
- b) TRUE. Gap = $29,045.86 - 28,750.00 = \295.86 , matching the statement exactly.
- c) FALSE. The actual ratio is $295.86 / 11.36 \approx 26.04$, which is more than 15 but less than 30 — the claim of "more than 30 times" overstates the true multiple.
- d) TRUE. Over 8 years at 3%, the gap grows to $31,781.23 - 31,669.25 = \111.98 , matching the statement exactly.
- e) FALSE. This directly contradicts both the calculation above and the textbook's stated rule that the difference between continuous and less-frequent compounding widens as the number of years increases: here the 8-year gap (\$111.98) is nearly ten times the 1-year gap (\$11.36), showing continuous compounding becomes MORE advantageous to the lender over longer holding periods, not less.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 10. Continuous Compounding as the Ceiling on a Fund's Return at an Investment Advisory Firm

Difficulty: 3.0/5

An investment advisory firm is evaluating a \$400,000 allocation and comparing two funds that both quote a nominal annual rate of 9.5%: Fund A compounds continuously, while Fund B compounds monthly. The firm also wants to check whether switching Fund B to daily compounding could ever let it catch up to or overtake Fund A.

Statements (True / False):

- a) Fund A's (continuous) year-end value is approximately \$439,863.54.
- b) Fund B's (monthly) year-end value is approximately \$439,750.00.
- c) The maximum possible effective annual rate obtainable at a 9.5% nominal rate, under any compounding frequency, is approximately 9.50% — the same as the nominal rate itself.
- d) If Fund B switched to daily compounding, its year-end value would exceed Fund A's continuous-compounding value.
- e) The dollar gap between Fund A and Fund B narrows when Fund B switches from monthly to daily compounding.

Solution – Task 10

Given:

- $P = \$400,000$
- Nominal annual rate $r = i = 9.5\% = 0.095$
- Fund A: continuous compounding
- Fund B: $m = 12$ (monthly); part d, e also consider $m = 365$ (daily)
- $t = 1$ year

Formula(s):

- $S_A = P \cdot e^{(rt)}$
- $S_B = P \cdot (1 + i/m)^{(mt)}$
- $EAR_{\max} = e^r - 1$

Step-by-Step Calculations:

Fund A: $S_A = 400,000 \times e^{0.095} = 400,000 \times 1.0996589 = \$439,863.54$.

Fund B (monthly): periodic rate $= 0.095/12 = 0.0079167$. $S_B = 400,000 \times (1.0079167)^{12} = 400,000 \times 1.0992476 = \$439,699.03$, not \$439,750.00.

$EAR_{\max} = e^{0.095} - 1 \approx 9.97\%$, which exceeds the 9.50% nominal rate; it is not equal to it.

Fund B (daily): $S_{\text{daily}} = 400,000 \times (1 + 0.095/365)^{365} = 400,000 \times 1.0996453 = \$439,858.10$, which is still below $S_A = \$439,863.54$.

Gap (continuous vs. monthly) $= 439,863.54 - 439,699.03 = \164.51 . Gap (continuous vs. daily) $= 439,863.54 - 439,858.10 = \5.44 .

Explanation of Each Statement:

- a) TRUE. $S_A = 400,000 \times e^{0.095} = \$439,863.54$, matching the statement exactly.
- b) FALSE. The correctly computed monthly-compounded value is $400,000 \times (1.0079167)^{12} = \$439,699.03$, not \$439,750.00 — the stated figure overstates the true monthly value by about \$50.97.
- c) FALSE. The continuous-compounding effective rate, $e^{0.095} - 1 \approx 9.97\%$, is the true ceiling on the effective annual rate at a 9.5% nominal rate, and it is noticeably higher than 9.50% — the nominal and effective rates are equal only in the special case of annual (once-per-year) compounding, not as a general ceiling.

d) FALSE. Daily compounding gives Fund B a value of approximately \$439,858.10, which remains below Fund A's continuous-compounding value of \$439,863.54, not above it. Since continuous compounding is the mathematical limit of compounding frequency as the number of periods tends to infinity, no finite compounding frequency — however frequent — can ever exceed it at the same nominal rate.

e) TRUE. The gap shrinks from \$164.51 (monthly) to \$5.44 (daily) as the discrete schedule compounds more often, illustrating exactly how $(1 + r/n)^n$ approaches e^r as n grows without ever reaching or surpassing it.

Final Answers — a: TRUE b: FALSE c: FALSE d: FALSE e: TRUE

Task 11. Reverse-Engineering the Implied Rate of a Boutique Winery's Futures Fund

Difficulty: 3.2/5

A boutique winery's grape futures fund grew continuously, at an unstated nominal annual rate, from \$28,000 to \$34,200 over the past 3 years. The fund manager wants to determine the implied rate and then project the fund's value 2 further years into the future (5 years from the start).

Statements (True / False):

- a) The implied nominal continuously-compounded rate is approximately 6.67%.
- b) Using this implied rate, the projected value 5 years from the start (2 years beyond the observed data) is approximately \$39,078.52.
- c) A naive straight-line projection — extending the average dollar increase observed over the first 3 years for 2 more years — gives the same result as the correct exponential projection.
- d) At the implied rate, the fund's value would double from its original \$28,000 in approximately 12.40 years.
- e) If the implied rate had instead been exactly 6.00%, the 3-year value would have been approximately \$33,522.09, which is higher than the actual observed \$34,200.00.

Solution – Task 11

Given:

- $S_0 = \$28,000$
- $S(3) = \$34,200$ (observed)
- Compounding: continuous
- $t = 3$ years (observed); project to $t = 5$ years

Formula(s):

- $S(t) = S_0 \cdot e^{(rt)} \Rightarrow r = \ln(S(t)/S_0)/t$
- Doubling time: $t = \ln(2)/r$

Step-by-Step Calculations:

$$r = \ln(34,200/28,000)/3 = \ln(1.221429)/3 = 0.200034/3 = 0.066674 \approx 6.67\%.$$

$$S(5) = 28,000 \times e^{(0.066674 \times 5)} = 28,000 \times e^{0.333368} = 28,000 \times 1.395661 = \$39,078.52.$$

Average dollar increase over 3 years = $(34,200 - 28,000)/3 = \$2,066.67/\text{year}$. Naive linear projection: $34,200 + 2 \times 2,066.67 = \$38,333.33$, which does not equal \$39,078.52.

$$\text{Doubling time} = \ln(2)/0.066674 = 0.693147/0.066674 \approx 10.3961 \text{ years.}$$

At $r = 6.00\%$: $S(3) = 28,000 \times e^{0.18} = 28,000 \times 1.197217 = \$33,522.09$, which is LOWER than the actual observed \$34,200.00, not higher.

Explanation of Each Statement:

- a) TRUE.** Dividing both sides of $S(3) = S_0 e^{(3r)}$ by S_0 and taking natural logs gives $3r = \ln(34,200/28,000)$, so $r = \ln(1.221429)/3 \approx 6.67\%$, matching the statement exactly.
- b) TRUE.** Applying the implied rate for a further 2 years (5 years total from the start) gives $S(5) = 28,000 \times e^{(0.066674 \times 5)} \approx \$39,078.52$, matching the statement exactly.
- c) FALSE.** The naive straight-line projection assumes the DOLLAR amount added each year stays constant, giving \$38,333.33, whereas the correct exponential projection assumes the PERCENTAGE growth rate stays constant, giving the higher figure of \$39,078.52. Because exponential growth compounds on an ever-larger base, the two methods diverge once projected beyond the observed data, and they do not match.
- d) FALSE.** $t = \ln(2)/0.066674 \approx 10.40$ years, not 12.40 years — the stated figure overstates the true doubling time by exactly 2 years.

e) FALSE. A rate of 6.00% instead of 6.67% would have produced a SMALLER 3-year value, approximately \$33,522.09, which is in fact lower than the actual observed \$34,200.00, not higher — a lower assumed growth rate can never explain a larger observed increase over the same period.

Final Answers — a: TRUE b: TRUE c: FALSE d: FALSE e: FALSE

Task 12. A Corporate Treasurer's Three-Bank Comparison With Differing Nominal Rates

Difficulty: 3.4/5

A corporate treasurer is placing \$60,000 for 2 years and compares three offers: Bank X compounds continuously at a nominal annual rate of 6.8%; Bank Y compounds monthly at a nominal annual rate of 6.9%; Bank Z compounds quarterly at a nominal annual rate of 7.0%.

Statements (True / False):

- a) Bank X's 2-year value is approximately \$68,740.91.
- b) Bank Y's 2-year value is approximately \$68,851.32.
- c) Bank Z's 2-year value is approximately \$68,932.91, making it the highest of the three offers.
- d) Despite compounding continuously, Bank X's value is actually the lowest of the three offers.
- e) If Bank X's nominal rate were instead also raised to 7.0%, while keeping continuous compounding, its 2-year value would exceed Bank Z's value.

Solution – Task 12

Given:

- $P = \$60,000$
- Bank X: $r = 6.8\% = 0.068$, continuous
- Bank Y: $i = 6.9\% = 0.069$, $m = 12$ (monthly)
- Bank Z: $i = 7.0\% = 0.070$, $m = 4$ (quarterly)
- $t = 2$ years

Formula(s):

- $S_X = P \cdot e^{(rt)}$
- $S_Y = P \cdot (1 + i/12)^{(12t)}$
- $S_Z = P \cdot (1 + i/4)^{(4t)}$

Step-by-Step Calculations:

$$S_X = 60,000 \times e^{(0.068 \times 2)} = 60,000 \times e^{0.136} = 60,000 \times 1.145682 = \$68,740.91.$$

$$S_Y = 60,000 \times (1 + 0.069/12)^{24} = 60,000 \times (1.00575)^{24} = 60,000 \times 1.147522 = \$68,851.32.$$

$$S_Z = 60,000 \times (1 + 0.070/4)^8 = 60,000 \times (1.0175)^8 = 60,000 \times 1.148882 = \$68,932.91.$$

Ordering the three results: $\$68,740.91$ (X) < $\$68,851.32$ (Y) < $\$68,932.91$ (Z), so Bank X is actually the LOWEST, not the highest.

If Bank X's rate were 7.0% instead of 6.8%: $S_X' = 60,000 \times e^{(0.070 \times 2)} = 60,000 \times e^{0.14} = 60,000 \times 1.150274 = \$69,016.44$, which does exceed Bank Z's $\$68,932.91$.

Explanation of Each Statement:

- a) TRUE. $S_X = 60,000 \times e^{0.136} = \$68,740.91$, matching the statement exactly.
- b) TRUE. $S_Y = 60,000 \times (1.00575)^{24} = \$68,851.32$, matching the statement exactly.
- c) TRUE. $S_Z = 60,000 \times (1.0175)^8 = \$68,932.91$, and comparing all three results confirms it is indeed the largest of the three.
- d) TRUE. The rule that continuous compounding is the most profitable schedule for the lender applies only when comparing different compounding FREQUENCIES at the SAME nominal rate. Here the three banks quote three DIFFERENT nominal rates, and Bank Z's higher quoted rate (7.0%) more than compensates for its less-frequent quarterly compounding, so Bank X's continuously-compounded value at the lowest rate (6.8%) does indeed end up the SMALLEST of the three, confirming the statement exactly.

e) TRUE. Once Bank X's nominal rate is raised to match Bank Z's 7.0%, the ceiling result does apply (same nominal rate, comparing continuous to quarterly), so continuous compounding at 7.0% (\$69,016.44) does exceed quarterly compounding at the same 7.0% (\$68,932.91).

Final Answers — a: TRUE b: TRUE c: TRUE d: TRUE e: TRUE

Task 13. Net Growth Rate of a Hedge Fund After a Continuous Management Fee Drag

Difficulty: 3.6/5

A hedge fund's gross asset value of \$2,000,000 grows continuously at a nominal annual rate of 9%, but the fund also deducts a continuous annual management fee of 2%, which acts as a constant drag on the growth rate.

Statements (True / False):

- a) The net continuous growth rate, combining the 9% gross return and the 2% fee drag, is 11% per year.
- b) After 6 years, the net asset value is approximately \$3,100,000.00.
- c) At this net rate, the fund's value would double in approximately 7.00 years.
- d) If the management fee instead rose to 3.5%, the 6-year net value would be approximately \$2,781,936.26, and the doubling time would shorten to approximately 12.60 years.
- e) A higher management fee always reduces both the net growth rate and the fund's cumulative value at any future date, compared to a lower fee, all else equal.

Solution – Task 13

Given:

- $S_0 = \$2,000,000$
- Gross rate = 9% = 0.09; fee = 2% = 0.02 (and 3.5% = 0.035 for part d)
- Compounding: continuous
- $t = 6$ years

Formula(s):

- Net rate: $r_{\text{net}} = r_{\text{gross}} - \text{fee}$
- $S(t) = S_0 \cdot e^{(r_{\text{net}} t)}$
- Doubling time: $t = \ln(2)/r_{\text{net}}$

Step-by-Step Calculations:

$$r_{\text{net}} = 0.09 - 0.02 = 0.07 = 7\%.$$

$$S(6) = 2,000,000 \times e^{(0.07 \times 6)} = 2,000,000 \times e^{0.42} = 2,000,000 \times 1.521962 = \$3,043,923.11.$$

$$\text{Doubling time} = \ln(2)/0.07 = 0.693147/0.07 \approx 9.9021 \text{ years.}$$

$$\text{At fee} = 3.5\%: r_{\text{net}}' = 0.09 - 0.035 = 0.055. S(6) = 2,000,000 \times e^{0.33} = 2,000,000 \times 1.390968 = \$2,781,936.26.$$

$$\text{Doubling time} = \ln(2)/0.055 \approx 12.6027 \text{ years, which is LONGER than 9.9021 years, not shorter.}$$

Comparing: a lower net rate (5.5% vs. 7%) always produces both a smaller 6-year value and a longer doubling time.

Explanation of Each Statement:

- a) FALSE.** The continuous fee drag must be SUBTRACTED from the gross rate, not added: $0.09 - 0.02 = 0.07$, giving a net rate of 7% per year, not 11%. Adding the fee to the gross rate would describe a fee that BOOSTS returns, which contradicts the very idea of a management fee reducing the investor's growth.
- b) FALSE.** The correctly computed net value is $S(6) = 2,000,000 \times e^{0.42} = \$3,043,923.11$, not \$3,100,000.00 — the stated figure overstates the true value by about \$56,076.89.
- c) FALSE.** The correct doubling time at the 7% net rate is $\ln(2)/0.07 \approx 9.90$ years, not 7.00 years; the stated figure understates the true doubling time by nearly 3 years.
- d) FALSE.** The 6-year value of \$2,781,936.26 at the higher 3.5% fee is correctly computed, but the doubling time at the resulting lower 5.5% net rate is $\ln(2)/0.055 \approx 12.60$ years, which is LONGER than the original 9.90 years, not shorter — a smaller net growth rate always takes MORE time to double an investment, never less.
- e) TRUE.** Since the net rate is gross rate minus fee, a higher fee mechanically produces a lower net rate, and a lower net continuous rate produces a smaller value at every future date $t > 0$ (because $e^{(rt)}$ is strictly increasing in r for fixed t).

> 0), so this general statement holds for any pair of fee levels compared under otherwise identical conditions.

Final Answers — a: FALSE b: FALSE c: FALSE d: FALSE e: TRUE

Task 14. Crossover Point Between a Growing Equity Stake and Shrinking Factory Equipment

Difficulty: 3.8/5

A boutique investment firm holds two assets: Asset A, a start-up equity stake currently worth \$50,000 and growing continuously at a nominal annual rate of 4%; and Asset B, aging factory equipment currently worth \$250,000 and depreciating continuously at an annual rate of 12%. The firm wants to know whether, and when, Asset A's value will overtake Asset B's value.

Statements (True / False):

- Setting $A_0 e^{(r_A t)} = B_0 e^{(-\delta_B t)}$ and solving for t gives $t = \ln(B_0/A_0)/(r_A + \delta_B)$.
- The crossover occurs at approximately $t \approx 10.06$ years, at which point both assets are worth approximately \$74,767.44.
- At exactly $t = 10$ years, Asset A is already worth more than Asset B.
- Asset A can never actually overtake Asset B in value, no matter how long both trends continue.
- For any time beyond the crossover point, Asset A's value remains higher than Asset B's value.

Solution – Task 14

Given:

- $A_0 = \$50,000$, $r_A = 4\% = 0.04$
- $B_0 = \$250,000$, $\delta_B = 12\% = 0.12$
- Compounding: continuous

Formula(s):

- $A(t) = A_0 \cdot e^{(r_A t)}$
- $B(t) = B_0 \cdot e^{(-\delta_B t)}$
- Crossover: $t = \ln(B_0/A_0)/(r_A + \delta_B)$

Step-by-Step Calculations:

$$t = \ln(250,000/50,000)/(0.04 + 0.12) = \ln(5)/0.16 = 1.609438/0.16 = 10.0590 \text{ years.}$$

$$A(10.0590) = 50,000 \times e^{(0.04 \times 10.0590)} = \$74,767.44. \quad B(10.0590) = 250,000 \times e^{(-0.12 \times 10.0590)} = \$74,767.44$$

(equal, as required at crossover).

$$A(10) = 50,000 \times e^{0.40} = 50,000 \times 1.491825 = \$74,591.23. \quad B(10) = 250,000 \times e^{(-1.20)} = 250,000 \times 0.301194 = \$75,298.55.$$

At $t = 10$, $A(10) = \$74,591.23 < B(10) = \$75,298.55$, so A has NOT yet overtaken B at that point — consistent with the crossover occurring slightly later, at 10.06 years.

Since $A_0 < B_0$, $A(t) < B(t)$ at $t = 0$, but because $A(t)/B(t) = (A_0/B_0) \cdot e^{((r_A + \delta_B)t)}$ grows without bound as t increases (a strictly increasing exponential ratio), the ratio must eventually exceed 1, guaranteeing a crossover exists.

Explanation of Each Statement:

- TRUE. Dividing both sides by $A_0 e^{(-\delta_B t)}$ and taking natural logs of the resulting equation $B_0/A_0 = e^{((r_A + \delta_B)t)}$ gives exactly $t = \ln(B_0/A_0)/(r_A + \delta_B)$, matching the statement exactly.
- TRUE. $t = \ln(5)/0.16 \approx 10.06$ years, and at that time both $A(t)$ and $B(t)$ equal approximately \$74,767.44, matching the statement exactly.
- FALSE. At $t = 10$ years, Asset A is worth approximately \$74,591.23 while Asset B is still worth approximately \$75,298.55 — Asset B is still ahead at this point, not Asset A, which is exactly why the crossover has not yet occurred by year 10 and instead happens slightly later, at about 10.06 years.
- FALSE. Although Asset B starts out five times larger, Asset A's slower but steady growth eventually overtakes Asset B's faster proportional shrinkage, because the RATIO $A(t)/B(t)$ grows exponentially without bound as t increases (at

combined rate $r_A + \delta_B = 16\%$ per year). No matter how large the initial gap, a strictly growing ratio must eventually cross 1, which is exactly what the calculated crossover time of about 10.06 years confirms.

e) TRUE. Once $A(t)$ exceeds $B(t)$ at the crossover, A continues to grow while B continues to shrink, so the gap between them ($A(t) - B(t)$) only widens further for all t beyond the crossover point, meaning Asset A remains ahead indefinitely thereafter.

Final Answers — a: TRUE b: TRUE c: FALSE d: FALSE e: TRUE

Task 15. Verifying the Compounding-Frequency Ceiling Across Four Schedules for a Municipal Reserve

Difficulty: 4.0/5

A municipal finance office is comparing four compounding schedules, all at the same nominal annual rate of 7%, applied to a \$40,000 reserve fund for exactly 1 year: annual, quarterly, monthly, and continuous. The office wants to confirm the textbook's claim that increasing compounding frequency strictly increases the year-end value, up to the continuous-compounding ceiling.

Statements (True / False):

- The annual-compounding value is exactly \$42,800.00.
- The quarterly-compounding value is approximately \$42,874.36, and the monthly-compounding value is approximately \$42,891.60.
- The four values, from smallest to largest, are ordered annual < quarterly < monthly < continuous.
- The dollar gap between monthly and quarterly compounding is smaller than the dollar gap between continuous and monthly compounding.
- No compounding schedule, however frequent, can ever produce a 1-year value exceeding the continuous-compounding value of \$42,900.33 at this same 7% nominal rate.

Solution – Task 15

Given:

- $P = \$40,000$
- Nominal annual rate $i = r = 7\% = 0.07$
- Schedules: annual ($m = 1$), quarterly ($m = 4$), monthly ($m = 12$), continuous
- $t = 1$ year

Formula(s):

- $S_m = P \cdot (1 + i/m)^m$ for finite m
- $S_{\text{cont}} = P \cdot e^r$

Step-by-Step Calculations:

Annual: $S = 40,000 \times 1.07 = \$42,800.00$.

Quarterly: $S = 40,000 \times (1.0175)^4 = 40,000 \times 1.0718590 = \$42,874.36$.

Monthly: $S = 40,000 \times (1.0058333)^{12} = 40,000 \times 1.0722901 = \$42,891.60$.

Continuous: $S = 40,000 \times e^{0.07} = 40,000 \times 1.0725082 = \$42,900.33$.

Gap(monthly, quarterly) = $42,891.60 - 42,874.36 = \17.24 . Gap(continuous, monthly) = $42,900.33 - 42,891.60 = \8.72 ; the second gap is SMALLER, not larger.

Explanation of Each Statement:

- TRUE. $40,000 \times 1.07 = \$42,800.00$, matching the statement exactly.
- TRUE. Quarterly gives \$42,874.36 and monthly gives \$42,891.60, and both do sit strictly between the annual value of \$42,800.00 and the continuous value of \$42,900.33, matching the statement exactly.
- TRUE. Listing the four results — \$42,800.00, \$42,874.36, \$42,891.60, \$42,900.33 — confirms they are strictly increasing in exactly this order as compounding frequency rises from annual through continuous, which is the textbook's claim being verified here.
- FALSE. The monthly-versus-quarterly gap is \$17.24, which is actually LARGER than the continuous-versus-monthly gap of \$8.72, not smaller. As compounding frequency keeps rising toward its continuous limit, each further increase in frequency contributes a progressively smaller additional dollar amount, since $(1 + r/n)^n$ is converging toward e^r and the increments shrink as n grows large.

e) TRUE. Since $(1 + r/n)^n \rightarrow e^r$ as $n \rightarrow \infty$, and the sequence is strictly increasing in n , the continuous-compounding value e^r represents the supremum (least upper bound) that no finite-frequency schedule at the same nominal rate can ever reach or exceed.

Final Answers — a: TRUE b: TRUE c: TRUE d: FALSE e: TRUE

Task 16. Reverse-Engineering the Upfront Deposit Needed for a Tuition Lock

Difficulty: 4.2/5

A parent wants to guarantee exactly \$100,000 available in 8 years for a child's college tuition, and is choosing between two continuously-compounded investment vehicles: Option 1 offers a nominal annual rate of 4.5%, while Option 2 offers a nominal annual rate of 6.0%. The parent wants to know how much must be deposited today under each option.

Statements (True / False):

- a) Option 1 requires an upfront deposit of approximately \$69,767.63.
- b) Option 2 requires an upfront deposit of approximately \$61,878.34.
- c) Option 2 requires a larger upfront deposit than Option 1 to reach the same \$100,000 target in 8 years.
- d) The difference in required upfront deposits between the two options is approximately \$9,000.00, with Option 2 requiring the larger amount.
- e) If the parent instead had only 4 years to reach the same \$100,000 target under Option 1's 4.5% rate, the required upfront deposit would be smaller than the 8-year requirement.

Solution – Task 16

Given:

- Target $S(t) = \$100,000$
- Option 1: $r = 4.5\% = 0.045$, $t = 8$ years (and $t = 4$ years for part e)
- Option 2: $r = 6.0\% = 0.06$, $t = 8$ years
- Compounding: continuous

Formula(s):

$$S(t) = S_0 \cdot e^{(rt)} \Rightarrow S_0 = S(t)/e^{(rt)} = S(t) \cdot e^{(-rt)}$$

Step-by-Step Calculations:

$$\text{Option 1: } S_0 = 100,000 \times e^{(-0.045 \times 8)} = 100,000 \times e^{(-0.36)} = 100,000 \times 0.697676 = \$69,767.63.$$

$$\text{Option 2: } S_0 = 100,000 \times e^{(-0.06 \times 8)} = 100,000 \times e^{(-0.48)} = 100,000 \times 0.618783 = \$61,878.34.$$

$$\text{Difference} = 69,767.63 - 61,878.34 = \$7,889.29, \text{ with Option 1 requiring more upfront.}$$

$$\text{Option 1 over 4 years: } S_0 = 100,000 \times e^{(-0.045 \times 4)} = 100,000 \times e^{(-0.18)} = 100,000 \times 0.835270 = \$83,527.02.$$

$$\text{Comparing: } \$83,527.02 \text{ (4-year requirement) is LARGER than } \$69,767.63 \text{ (8-year requirement), not smaller.}$$

Explanation of Each Statement:

- a) TRUE.** $S_0 = 100,000 \times e^{(-0.36)} = \$69,767.63$, matching the statement exactly.
- b) TRUE.** $S_0 = 100,000 \times e^{(-0.48)} = \$61,878.34$, matching the statement exactly.
- c) FALSE.** This has the direction backwards: a higher growth rate means each dollar deposited today grows to a larger sum by year 8, so REACHING the same \$100,000 target actually requires depositing LESS upfront under the higher, 6.0% rate (\$61,878.34) than under the lower, 4.5% rate (\$69,767.63), not more.
- d) FALSE.** The correct difference is $69,767.63 - 61,878.34 = \$7,889.29$, not \$9,000.00, and it is Option 1 (the LOWER rate) that requires the larger upfront amount, not Option 2 — the statement gets both the figure and the direction wrong.
- e) FALSE.** With less time available to grow (4 years instead of 8), MORE money must be set aside today to still reach the same \$100,000 target, not less. The correct 4-year requirement is \$83,527.02, which is larger than the 8-year requirement of \$69,767.63, since a shorter compounding horizon leaves less room for exponential growth to do the work.

Final Answers — a: TRUE b: TRUE c: FALSE d: FALSE e: FALSE

Task 17. A Logistics Company's Two-Phase Continuous Growth: Expansion Then Maturity

Difficulty: 4.4/5

A logistics company's revenue base of \$1,800,000 grows continuously at a nominal annual rate of 10% for the first 4 years (a rapid expansion phase), then slows to a continuous nominal rate of 4% for the following 3 years (a maturity phase), for 7 years total.

Statements (True / False):

- Revenue at the end of year 4 is approximately \$2,685,284.46.
- Revenue at the end of year 7 is approximately \$3,027,649.77.
- The single constant continuous rate that would have produced the same 7-year outcome starting from \$1,800,000 is approximately 7.43%.
- The effective 7-year rate is higher than the plain (unweighted) average of the two phase rates.
- If the two phases had instead occurred in the opposite order — 3 years at 4% followed by 4 years at 10% — the year-7 revenue would have come out exactly the same.

Solution – Task 17

Given:

- $S_0 = \$1,800,000$
- Phase 1: $r_1 = 10\% = 0.10$, $t_1 = 4$ years
- Phase 2: $r_2 = 4\% = 0.04$, $t_2 = 3$ years

Formula(s):

- $S(t_1) = S_0 \cdot e^{(r_1 t_1)}$
- $S(t_1 + t_2) = S(t_1) \cdot e^{(r_2 t_2)} = S_0 \cdot e^{(r_1 t_1 + r_2 t_2)}$
- Effective rate: $r_{\text{eff}} = (r_1 t_1 + r_2 t_2) / (t_1 + t_2)$

Step-by-Step Calculations:

$$S(4) = 1,800,000 \times e^{(0.10 \times 4)} = 1,800,000 \times e^{0.40} = 1,800,000 \times 1.491825 = \$2,685,284.46.$$

$$S(7) = S(4) \times e^{(0.04 \times 3)} = 2,685,284.46 \times e^{0.12} = 2,685,284.46 \times 1.127497 = \$3,027,649.77.$$

$$\text{Combined exponent} = r_1 t_1 + r_2 t_2 = (0.10 \times 4) + (0.04 \times 3) = 0.40 + 0.12 = 0.52.$$

$$r_{\text{eff}} = 0.52 / 7 = 0.074286 \approx 7.43\% \text{ (the TIME-WEIGHTED average, not the plain average).}$$

$$\text{Plain (unweighted) average} = (0.10 + 0.04) / 2 = 0.07 = 7.00\%, \text{ which does not equal } 7.43\%.$$

Explanation of Each Statement:

- TRUE. $S(4) = 1,800,000 \times e^{0.40} = \$2,685,284.46$, matching the statement exactly.
- TRUE. Applying the second phase's factor to the year-4 balance gives $S(7) = 2,685,284.46 \times e^{0.12} = \$3,027,649.77$, matching the statement exactly.
- TRUE. Because $S(7) = S_0 \cdot e^{(r_1 t_1 + r_2 t_2)}$, the single constant rate reproducing the same 7-year result solves $r_{\text{eff}} = (r_1 t_1 + r_2 t_2) / 7 = 0.52 / 7 \approx 7.43\%$, matching the statement exactly.
- TRUE. The correct effective rate is the TIME-WEIGHTED average of the two phase rates, weighted by how many years each phase lasted: $(10\% \times 4 + 4\% \times 3) / 7 = 7.43\%$, which is indeed higher than the plain, unweighted average of $(10\% + 4\%) / 2 = 7.00\%$. This makes sense because the faster-growing 10% phase lasted longer (4 years vs. 3 years) and so should count for more, pulling the true effective rate above the simple average.
- TRUE. Since $S_0 \cdot e^{(r_1 t_1)} \cdot e^{(r_2 t_2)} = S_0 \cdot e^{(r_1 t_1 + r_2 t_2)}$, and ordinary multiplication is commutative, the combined exponent $r_1 t_1 + r_2 t_2 = 0.52$ is exactly the same regardless of which phase happens first. Reversing the order of the two phases therefore produces the identical year-7 revenue of \$3,027,649.77, exactly as stated.

Final Answers — a: TRUE b: TRUE c: TRUE d: TRUE e: TRUE

Task 18. Reverse-Engineering a Crane's Implied Depreciation Rate for a Construction Equipment Reseller

Difficulty: 4.6/5

A construction equipment reseller has a crane currently valued at \$85,000 that, per accounting policy, must be written down to a resale value of \$32,000 after 6 years of continuous depreciation. The reseller wants to know the implied annual depreciation rate δ , and compare the crane's value retention to a second, otherwise identical crane with a known $\delta = 15\%$.

Statements (True / False):

- Solving $v_0 \cdot e^{(-\delta t)} = v(t)$ for δ gives $\delta = \ln(v(t)/v_0)/t$.
- The implied depreciation rate for the first crane is approximately 16.28% per year.
- A second, otherwise identical crane purchased for the same \$85,000 but depreciating continuously at a known rate of 15% per year would be worth approximately \$36,000.00 after the same 6 years.
- The first crane retains more of its value after 6 years than the second crane.
- If the first crane's target resale value had instead been set at \$40,000 after the same 6 years, the implied depreciation rate would be higher than 16.28%.

Solution – Task 18

Given:

- $v_0 = \$85,000$
- $v(6) = \$32,000$ (first crane, implied δ); known $\delta = 15\%$ (second crane); $v(6)$ target = \$40,000 for part e
- Compounding: continuous
- $t = 6$ years

Formula(s):

$$v(t) = v_0 \cdot e^{(-\delta t)} \Rightarrow \delta = \ln(v_0/v(t))/t$$

Step-by-Step Calculations:

$$\delta = \ln(85,000/32,000)/6 = \ln(2.65625)/6 = 0.976915/6 = 0.162819 \approx 16.28\%.$$

$$\text{Second crane: } v(6) = 85,000 \times e^{(-0.15 \times 6)} = 85,000 \times e^{(-0.90)} = 85,000 \times 0.406570 = \$34,558.42.$$

Comparing final values: first crane retains \$32,000.00, second crane retains \$34,558.42 — the crane with the LOWER rate (15%) retains MORE value, not the one with the higher implied rate (16.28%).

For a \$40,000 target: $\delta = \ln(85,000/40,000)/6 = \ln(2.125)/6 = 0.753772/6 = 0.125629 \approx 12.56\%$, which is lower than 16.28%.

Explanation of Each Statement:

- FALSE. This has the ratio inverted: dividing both sides by v_0 and taking natural logs of $v_0/v(t)$ — not $v(t)/v_0$ — gives $\delta t = \ln(v_0/v(t))$, so the correct formula is $\delta = \ln(v_0/v(t))/t$. Using $\ln(v(t)/v_0)/t$ instead would produce a NEGATIVE number here, since $v(t) < v_0$, which cannot be a valid depreciation rate.
- TRUE. $\delta = \ln(85,000/32,000)/6 \approx 16.28\%$, matching the statement exactly.
- FALSE. The correctly computed value is $v(6) = 85,000 \times e^{(-0.90)} = \$34,558.42$, not \$36,000.00 — the stated figure overstates the true value by about \$1,441.58.
- FALSE. This has the logic backwards: a HIGHER depreciation rate means the asset loses value FASTER, not slower, so the first crane (implied rate 16.28%) actually retains LESS value after 6 years (\$32,000.00) than the second crane at the lower 15% rate (\$34,558.42), not more.
- FALSE. A higher target resale value of \$40,000 (instead of \$32,000) means LESS value was lost over the same 6 years, which corresponds to a SMALLER implied depreciation rate, not a larger one: $\delta = \ln(85,000/40,000)/6 \approx 12.56\%$, which is LOWER than the original 16.28%, contradicting the statement.

Final Answers — a: FALSE b: TRUE c: FALSE d: FALSE e: FALSE

Task 19. Doubling, Tripling, and Quadrupling Times for an Impact Investing Fund

Difficulty: 4.8/5

A socially-responsible impact investing fund places \$12,000 into a continuously-compounded account at a nominal annual rate of 6.5%, and the fund's trustees want to understand exactly how the times needed to double, triple, and quadruple the initial investment relate to one another.

Statements (True / False):

- a) The doubling time at 6.5% is approximately 10.66 years.
- b) The tripling time at 6.5% is approximately 16.90 years.
- c) The quadrupling time at 6.5% is approximately 21.33 years, and this is exactly equal to twice the doubling time.
- d) At the quadrupling time, the fund's value is exactly \$48,000.00.
- e) The tripling time must be exactly 1.5 times the doubling time.

Solution – Task 19

Given:

- $S_0 = \$12,000$
- Nominal annual rate $r = 6.5\% = 0.065$
- Compounding: continuous
- Targets: $S(t) = 2S_0, 3S_0, 4S_0$

Formula(s):

- $S(t) = S_0 \cdot e^{(rt)}$
- Time to reach multiple M : $t = \ln(M)/r$

Step-by-Step Calculations:

Doubling: $t = \ln(2)/0.065 = 0.693147/0.065 \approx 10.6638$ years.

Tripling: $t = \ln(3)/0.065 = 1.098612/0.065 \approx 16.9017$ years.

Quadrupling: $t = \ln(4)/0.065 = 1.386294/0.065 \approx 21.3276$ years.

$2 \times (\text{doubling time}) = 2 \times 10.6638 = 21.3276$ years, which exactly equals the quadrupling time, since $\ln(4) = \ln(2^2) = 2 \cdot \ln(2)$.

Ratio of tripling time to doubling time $= \ln(3)/\ln(2) \approx 1.5850$, not 1.5; value at quadrupling time $= 12,000 \times 4 = \$48,000.00$.

Explanation of Each Statement:

- a) TRUE. $t = \ln(2)/0.065 \approx 10.66$ years, matching the statement exactly.
- b) TRUE. $t = \ln(3)/0.065 \approx 16.90$ years, matching the statement exactly.
- c) TRUE. $t = \ln(4)/0.065 \approx 21.33$ years, and because $4 = 2^2$ means $\ln(4) = 2\ln(2)$, the quadrupling time is exactly twice the doubling time — reaching four times the principal is mathematically the same as doubling the investment twice in a row.
- d) TRUE. Four times the original \$12,000 principal is $12,000 \times 4 = \$48,000.00$, matching the statement exactly.
- e) FALSE. Times to reach a given multiple scale with the NATURAL LOGARITHM of that multiple, not with the multiple itself, since $t = \ln(M)/r$. The ratio of the tripling time to the doubling time is therefore $\ln(3)/\ln(2) \approx 1.5850$, not the naive ratio of the multipliers themselves ($3/2 = 1.5$) — these two ratios are close but not equal, and treating them as the same is a common but incorrect shortcut.

Final Answers — a: TRUE b: TRUE c: TRUE d: TRUE e: FALSE

Task 20. Capstone: A Three-Asset Family Office Portfolio Under Continuous Growth and Decay

Difficulty: 5.0/5

A family office's capstone valuation combines three holdings, all valued in nominal (undiscounted) dollars and added together just like ordinary fixed amounts. Asset A is a private equity stake currently worth \$150,000, growing continuously at a nominal annual rate of 6% for 5 years. Asset B is aging warehouse machinery currently worth \$220,000, depreciating continuously at an annual rate of 9% for the same 5 years. Asset C is a licensing agreement currently worth \$100,000 that grows continuously at 8% for its first 3 years before slowing to a continuous 3% for its remaining 2 years (5 years total).

Statements (True / False):

- Asset A's value after 5 years is approximately \$202,478.82.
- Asset B's value after 5 years is approximately \$140,278.19.
- Asset C's value after 5 years is approximately \$130,000.00.
- The combined portfolio value after 5 years, adding all three assets together, is approximately \$477,742.90, which is less than the sum of the three original principals.
- If Asset B's rate had instead been a continuous growth rate of the same 9% magnitude, its 5-year value would exceed \$340,000.00.

Solution – Task 20

Given:

- Asset A: $A_0 = \$150,000$, $r_A = 6\% = 0.06$, $t = 5$ years
- Asset B: $B_0 = \$220,000$, $\delta_B = 9\% = 0.09$, $t = 5$ years
- Asset C: $C_0 = \$100,000$, $r_1 = 8\%$ for 3 years, then $r_2 = 3\%$ for 2 years

Formula(s):

- $A(t) = A_0 \cdot e^{(r_A t)}$
- $B(t) = B_0 \cdot e^{(-\delta_B t)}$
- $C(5) = C_0 \cdot e^{(r_1 \cdot 3)} \cdot e^{(r_2 \cdot 2)}$
- Portfolio total = $A(5) + B(5) + C(5)$

Step-by-Step Calculations:

$$A(5) = 150,000 \times e^{(0.06 \times 5)} = 150,000 \times e^{0.30} = 150,000 \times 1.349859 = \$202,478.82.$$

$$B(5) = 220,000 \times e^{(-0.09 \times 5)} = 220,000 \times e^{(-0.45)} = 220,000 \times 0.637628 = \$140,278.19.$$

$$C(3) = 100,000 \times e^{(0.08 \times 3)} = 100,000 \times e^{0.24} = 100,000 \times 1.271249 = \$127,124.92. \quad C(5) = 127,124.92 \times e^{(0.03 \times 2)} = 127,124.92 \times e^{0.06} = 127,124.92 \times 1.061837 = \$134,985.88.$$

Portfolio total = $202,478.82 + 140,278.19 + 134,985.88 = \$477,742.89$ (rounding to the cent). Sum of principals = $150,000 + 220,000 + 100,000 = \$470,000.00$; the total (\$477,742.89) is MORE than the principal sum, not less.

If B0 grew instead of shrinking at 9%: $B(5)' = 220,000 \times e^{(0.09 \times 5)} = 220,000 \times e^{0.45} = 220,000 \times 1.568312 = \$345,028.68$, which does exceed \$340,000.00.

Explanation of Each Statement:

- TRUE. $A(5) = 150,000 \times e^{0.30} = \$202,478.82$, matching the statement exactly.
- TRUE. $B(5) = 220,000 \times e^{(-0.45)} = \$140,278.19$, matching the statement exactly.
- FALSE. Chaining the two growth phases correctly gives $C(3) = \$127,124.92$ and then $C(5) = C(3) \times e^{0.06} = \$134,985.88$, not \$130,000.00 — the stated figure understates the true value by about \$4,985.88.
- FALSE. Adding the three 5-year values gives $202,478.82 + 140,278.19 + 134,985.88 = \$477,742.89$, which is actually MORE than the \$470,000.00 sum of the three original principals, not less — the combined growth of Assets A

and C outweighs the shrinkage of Asset B over this particular 5-year horizon.

e) TRUE. Replacing Asset B's 9% depreciation with a 9% continuous growth rate over the same 5 years gives $220,000 \times e^{0.45} \approx \$345,028.68$, which does exceed \$340,000.00, illustrating just how large the swing is between shrinking and growing at the same magnitude rate over a multi-year horizon.

Final Answers — a: TRUE b: TRUE c: FALSE d: FALSE e: TRUE

This worksheet (Tasks 1–20) relies exclusively on the continuous-compounding formula $S(t) = S_0 e^{rt}$, the fixed yearly growth factor e^r , the effective-rate formula $e^r - 1$ as the ceiling over any discrete compounding frequency at the same nominal rate, the general rule that continuous compounding is the most profitable schedule for the lender (with the gap widening as the rate or horizon grows), and the continuous-depreciation formula $v(t) = v_0 e^{-\delta t}$, all combined in multi-step ways across Tasks 11–20, exactly as presented in Section 11.2.